

Role of insulin in the pathogenesis of free fatty acid-induced insulin resistance in skeletal muscle.

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Insulin resistance is a pathophysiological link of obesity to type 2 diabetes. The initial cause of insulin resistance is critical for prevention and treatment of type 2 diabetes. Lipotoxicity is a well-known concept in the explanation of initiation of insulin resistance. Although there are several prevailing hypotheses about the cellular/molecular mechanisms of lipotoxicity, such as inflammation, oxidative stress, hyperinsulinemia, and ER stress, the relative importance of these hypothesized events remains to be determined. The role of hyperinsulinemia is relatively under documented in the literature for the initiation of insulin resistance. In this review, an interaction of fatty acid and beta-cells, and a synergy between free fatty acids (FFAs) and insulin are emphasized for the role of hyperinsulinemia. This article presents the evidence about FFA-induced insulin secretion in vitro and in vivo, recent advances in the molecular mechanism of FFA action in beta-cells, a role of GPR40 in the development of insulin resistance, and the negative feedback loop of the insulin receptor signal pathway. The negative feedback loop is discussed in detail with a focus on IRS-1 serine kinases. This article provides a substantial support for the role of insulin in the early stages of FFA-associated insulin resistance. The hypothesis of insulin's role in lipotoxicity is referred to as the "insulin hypothesis" in this review. According to this hypothesis, prevention of increased beta-cell response to glucose may be a potential approach for early intervention of metabolic syndrome.